Old SolarSizer vs Current SDK2: Feature Comparison

Old repo: `C:\Users\rotim\AppData\Local\Temp\opencode\solar-remix` (references below use `old/...`) **Current repo:** `C:\Users\rotim\Documents\Solar_Sizer_SDK2\Solar_Sizer_SDK2-9c02b14371d2830807c357d2bb5473dce5d658c7` (references below use `cur/...`)

Snapshot: 2026-11-09. Current `src/App.tsx` has uncommitted working-tree changes on top of `6ccd8fe` (+160/−66 lines), including a Bachs checkout button, a redesigned system-details modal, overnight-range fixes in the usage profile and saved-profile PDF, and the restored `InteractiveBridge onChange` handler. `worker/index.ts` is also modified (+53 lines) with a new `/api/checkout` handler. This report describes that working tree, not the last commit; all other current files match `6ccd8fe`.

Method: both trees were read end to end, not mechanically diffed. `App.tsx` was read in full on both sides (old 3,192 lines, current 4,636 lines), plus both calculators, both API layers, types, constants, auth components, build configs, migrations, seeds, and export functions.

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1. Executive summary: top 10 ranked

#	Finding	Old behaviour	Current behaviour
1	**User accounts and server-side persistence are gone.**	Google OAuth popup, session cookie, per-user profiles/results/hardware in Supabase or SQLite (`old/server.ts:101-239`, `old/server.ts:241-376`, `old/src/components/Auth.tsx:1-77`).	Auth component is an 18-line stub that always reports `null` (`cur/src/components/Auth.tsx:1-18`); the Worker keeps the old URLs but returns empty arrays and silently accepts/discards writes (`cur/worker/index.ts:238-260`). Profiles/results/hardware live only in `localStorage` (`cur/src/App.tsx:610-612`, `cur/src/App.tsx:964-989`, `cur/src/App.tsx:1002-1021`). Clearing browser storage destroys all user work.
2	**Local custom hardware is overwritten by global hardware on every load.**	Server hardware was merged into local arrays by id, preserving local-only items (`old/src/App.tsx:511-549`).	Any non-empty global list replaces the whole local array (`cur/src/App.tsx:674-683`), and the persistence effect then writes the replacement back to `localStorage` (`cur/src/App.tsx:955-962`). A non-admin user's added inverter/panel/battery/powerstation is lost on refresh.
3	**High Risk systems no longer reach the UI.**	All statuses were pushed, including High Risk (`old/src/utils/solarCalcul 411`).	The engine skips every High Risk result (`cur/src/utils/solarCalculator.ts:336`, `cur/src/utils/solarCalculator.ts:706-709`). The UI still contains High Risk branches that are now unreachable (`cur/src/App.tsx:2679`, `cur/src/App.tsx:2725`, `cur/src/App.tsx:4148-4156`), and users whose load needs an undersized system get "No Matching Systems" instead of a warned option.
	Checko exists		`startBachsCheckout` posts to `/api/checkout` (`cur/src/App.tsx:2155-2178`, button at

4	**but trusts the client's price for custom systems**	No checkout.	`cur/src/App.tsx:4311-4318`). The Worker route (`cur/worker/index.ts:399-448`) looks up catalog prices from D1 but accepts `body.amount` for custom systems. It also needs an undocumented `BACHS_API_KEY` (missing from `cur/.env.example:1-7`) and returns to `?payment=...` URLs that no UI code reads (`cur/worker/index.ts:435-436`).
5	**Compar PDF has a column mismat and hides the generat purchas cost.**	Rows and headers both always appended the generator, and `initialCost` was `hasGenerator ? 0 : selectedGen.price`, showing "Owned" or a price (`old/src/App.tsx:206`, `old/src/App.tsx:224-235`).	Header appends "Generator" only when `hasGen` is true (`cur/src/App.tsx:1133`); rows 1137-1142 append a generator cell unconditionally while the Initial Cost row at 1136 never does, so the table is misaligned either way. `initialCost` is hard-coded to `0` (`cur/src/App.tsx:1127`), so the generator is always treated as owned even when `has_generator` is false (`cur/src/App.tsx:3849`).
6	**Powerst are missing from the hardwa exports**	Feature did not exist.	JSON exports `{ inverters, panels, batteries }` only (`cur/src/App.tsx:1380-1384`). Markdown export has sections 1-3 only (`cur/src/App.tsx:1726-1751`). Import accepts `powerstations` (`cur/src/App.tsx:1472-1483`), so export then import loses them.
7	**Admin is one shared passwo stored in `sessio`, with no throttli and client-supplie catalog data.**	Google identity plus a developer email allow-list (`old/src/App.tsx:499-500`).	`?admin=true` prompts once, posts to `/api/admin/verify`, and stores the raw password in `sessionStorage` (`cur/src/App.tsx:635-663`). The Worker compares it timing-safel (`cur/worker/index.ts:56-73`, `cur/worker/index.ts:226-236`) but has no rate limit, lockout, or per-user role. Product writes tru the client payload, including price (`cur/worker/index.ts:339-36`).
8	**"Peak Load" in the results PDF is not a peak.**	Results PDF listed surge, nightly, daily, and count only (`old/src/App.tsx:967-1029`).	New row computes `Math.max(...Object.values(results.analysis.hourly_c` (`cur/src/App.tsx:1246-1250`), which is the highest hourly average, not instantaneous peak or surge (surge is the next row). Misleading label on a customer-facing report.
9	**Dev and deploy now require**	`npm run dev` started Express plus Vite in one process (`old/package.json:6`,	`npm run dev` starts Vite only, and `/api` is proxied to a separate Worker on port 8787 (`cur/vite.config.ts:36-38`, `cur/package.json:6-15`). `npm start` and `dev:worker` start

	two runtime plus D1.	old/server.ts:384-388).	Wrangler (cur/package.json:10-11), so local work needs two terminals unless the static build is served by Wrangler.
10	**Dead code and duplicat logic hide real drift.**	Fewer layers, one comparison PDF implementation (old/src/App.tsx:212-247).	exportSingleResultToPDF (cur/src/App.tsx:1171) and exportProfileToPDF (cur/src/App.tsx:1291) are defined and never called, as in the old tree (old/src/App.tsx:914, old/src/App.tsx:1031). New dead helpers: getStatusFromSimulation (cur/src/utils/solarCalculator.ts:208) is never called; the comparison modal recomputes generator maths (cur/src/App.tsx:337-355) that the PDF function recomputes again (cur/src/App.tsx:1119-1130).

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2. Feature matrix

Legend: "same" means behaviour and implementation are materially unchanged; "moved" means the feature exists but its storage or location changed; "lost" means the old capability has no working equivalent.

2.1 Navigation and screens

Feature	Old	Current	Verdict
Tab model	calculator | database | logs | profiles | results (old/src/types.ts:123)	calculator | products | internet | database | logs | profiles | results (cur/src/types.ts:214)	Expanded
Calculator tab	Always available (old/src/App.tsx:1562-1567)	Always available (cur/src/App.tsx:2220-2225)	Same
Profiles tab	Visible only when signed in (old/src/App.tsx:1568-1575)	Always visible (cur/src/App.tsx:2238-2243)	Access opened, backend lost
Results tab	Visible only when signed in (old/src/App.tsx:1576-1581)	Always visible (cur/src/App.tsx:2244-2250)	Access opened, backend lost
Hardware DB tab	Visible only when signed in (old/src/App.tsx:1582-1587)	Always visible (cur/src/App.tsx:2251-2256)	Access opened
Logs tab	Signed-in developer emails only (old/src/App.tsx:499-500, old/src/App.tsx:1588-1595, old/src/App.tsx:2281)	Admin unlock only (cur/src/App.tsx:2256-2264, cur/src/App.tsx:3515)	Different gate, same hidden-by-default result
Products tab	Does not exist	New catalog with tag filters, admin import/export, price display (cur/src/App.tsx:2910-3074)	New
Internet tab	Does not exist	New Starlink/router catalog (cur/src/App.tsx:3074-3190)	New

Header auth widget	Google sign-in button, avatar dropdown, Sign Out, Settings "Soon" item (`old/src/components/Auth.tsx:96-215`)	Removed; no auth widget (header runs `cur/src/App.tsx:2208-2278`)	Lost
Compact / URL-param mode	Does not exist	`?tab=`, `?tag=`, `?compact=true` handled on mount (`cur/src/App.tsx:617-627`, header hidden at `cur/src/App.tsx:2209`)	New
Mobile nav	`hidden md:flex`; no mobile replacement (`old/src/App.tsx:1561`)	`hidden md:flex`; no mobile replacement (`cur/src/App.tsx:2219`)	Same gap

2.2 Calculator workflow

Feature	Old	Current	Verdict
Region selector	`SE_SS`, `SW`, `North` (`old/src/App.tsx:1623-1640`)	Same (`cur/src/App.tsx:2285-2312`)	Same
Battery preference	any / lithium / lead-acid (`old/src/App.tsx:1642-1665`)	Same (`cur/src/App.tsx:2313-2335`)	Same
Tolerance slider	0-50%, default 20 (`old/src/App.tsx:1667-1690`)	Same (`cur/src/App.tsx:2336-2360`)	Same
Device add form	Name, category, qty, watts, multiple time ranges (`old/src/App.tsx:1343-1394`)	Same, plus powerstation category support in modal (`cur/src/App.tsx:1765-1834`, `cur/src/App.tsx:4490-4535`)	Same plus extensic
Time-range wrap	Engine handles overnight wrap (`old/src/utils/solarCalculator.ts:33`)	Engine still handles wrap (`cur/src/utils/solarCalculator.ts:33`); PDFs now also wrap (working tree `cur/src/App.tsx:1973`, `cur/src/App.tsx:3661`)	Same, PDF bug fixed in working tree
Master device quick-add	Does not exist	Add from `devices_master` list into the load (`cur/src/App.tsx:2372-2400`)	New
Save profile	Requires sign-in; POST `/api/user/profiles` (`old/src/App.tsx:710-757`)	`localStorage` only (`cur/src/App.tsx:964-993`)	Backenc lost
Save single result	Requires sign-in; POST `/api/user/results` (`old/src/App.tsx:774-806`)	`localStorage` only (`cur/src/App.tsx:1002-1021`)	Backenc lost
Save analysis	Requires sign-in (`old/src/App.tsx:808-843`)	`localStorage` only (`cur/src/App.tsx:1024-1047`)	Backenc lost
Save comparison	Requires sign-in (`old/src/App.tsx:845-884`)	`localStorage` only (`cur/src/App.tsx:1049-1074`)	Backenc lost
Delete result	Requires sign-in; DELETE `/api/user/results/:id` (`old/src/App.tsx:886-895`)	Local array removal (`cur/src/App.tsx:1076-1082`)	Backenc lost
Result ordering	Optimal → Conditional → High Risk, then price (`old/src/App.tsx:673-680`)	Same sort (`cur/src/App.tsx:903-910`), but High Risk rows never exist	Drift
Result category	Does not exist	Full Results / Kits / Powerstations (`cur/src/App.tsx:2605-2628`, filter logic `cur/src/App.tsx:2664-2667`	New

filter		at)	
Result pagination	Renders all systems	10 per page (`cur/src/App.tsx:606-607`, `cur/src/App.tsx:2691-2693`, controls `cur/src/App.tsx:2822-2855`)	New
"Cheapest of each category first"	Does not exist	Pins cheapest Optimal, Conditional, High Risk before the rest (`cur/src/App.tsx:2678-2688`)	New
Result detail modal	Separate panels for optimal/conditional/high risk (`old/src/App.tsx:2892-2960`)	Redesigned modal, status banner, three spec cards, Buy button (working tree `cur/src/App.tsx:4098-4324`)	Redesig
InteractiveBri	Conditional systems only; `onChange` writes `adjustedLoad` (`old/src/App.tsx:2933-2941`)	Working tree restored to Conditional only with `onChange` (`cur/src/App.tsx:4160-4178`); committed HEAD version had no `onChange` and also offered High Risk	Restore in working tree
Comparison selection	Checkbox per system, sticky comparison bar, modal (`old/src/App.tsx:614-642`, `old/src/App.tsx:2498-2560`)	Same pattern (`cur/src/App.tsx:841-874`, `cur/src/App.tsx:2856-2890`)	Same
Generator comparison	`has_generator` toggle persisted to localStorage (`old/src/App.tsx:604-610`)	Same (`cur/src/App.tsx:831-837`)	Same
Social/marke footer	Does not exist	Footer with links and newsletter form (`cur/src/App.tsx:4053-4100`)	New

2.3 Profiles, results, database, logs

Feature	Old	Current	Verdict
Profiles list	Search, load, PDF export, delete (`old/src/App.tsx:2365-2498`)	Search, load, JSON export/import, delete (`cur/src/App.tsx:3612-3736`)	PDF replaced by JSON
Profiles inventory PDF	Inline autotable in the profiles tab (`old/src/App.tsx:2404-2428`)	No inline PDF; `exportProfileToPDF` exists but is never called (`cur/src/App.tsx:1291`)	Lost
Saved results list	Server-backed; view, PDF, delete (`old/src/App.tsx:2498-2831`)	`localStorage`-backed; view, PDF, delete, filters (`cur/src/App.tsx:3736-4053`)	Storage moved
Saved comparison view	Loads `savedResults` with `is_comparison` (`old/src/App.tsx:2590-2645`)	Same plus stored `devices` snapshot (`cur/src/App.tsx:3830-3870`, `cur/src/types.ts:167-177`)	Improved
Hardware DB sections	Inverters, panels, batteries (`old/src/App.tsx:2148-2281`)	Inverters, panels, batteries, powerstations (`cur/src/App.tsx:3190-3515`)	Expanded
Master device CRUD	Does not exist	Add/edit/delete/import in admin panel (`cur/src/App.tsx:3330-3420`, `cur/src/App.tsx:4328-4420`)	New
	Local arrays plus fire-and-forget POST when signed in (	Local arrays; admin edits also write to D1 (	

Hardware CRUD	old/src/App.tsx:644-655, old/src/App.tsx:1413-1444)	cur/src/App.tsx:1835-1851, cur/src/App.tsx:4455-4492)	Split source of truth
Product CRUD	Does not exist	Admin save/delete/import/export (cur/src/App.tsx:751-826, cur/src/App.tsx:1538-1566)	New
Profit margins	Does not exist	Per-component percentages applied in engine (cur/src/App.tsx:3247-3270, cur/src/utils/solarCalculator.ts:331, cur/src/utils/solarCalculator.ts:409-417)	New
PDF quote customization	Hard-coded title, prefix, footers (old/src/App.tsx:1448, old/src/App.tsx:1463, old/src/App.tsx:1520-1521)	Editable title, prefix, two footer lines (cur/src/App.tsx:3278-3345, consumed at cur/src/App.tsx:2037, cur/src/App.tsx:2059, cur/src/App.tsx:2145-2146)	New
Logs capture	Inline inside useMemo, cap 50, duplicates possible (old/src/App.tsx:682-697)	useEffect, duplicate suppression, cap 20 (cur/src/App.tsx:916-946)	Improved
Logs clear	Does not exist	"Clear All Logs" with confirm (cur/src/App.tsx:3532-3543)	New
Log detail	Expandable traces (old/src/App.tsx:2281-2365)	Same plus per-log TXT export (cur/src/App.tsx:3515-3612)	Same
Storage quota guard	None; raw localStorage.setItem	safeLocalStorageSet catches quota errors (cur/src/App.tsx:878-890)	New

2.4 Settings and admin

Feature	Old	Current	Verdict
Settings entry	Disabled "Settings (Soon)" menu item (old/src/components/Auth.tsx:150-158)	No user settings menu; admin panels inside tabs	Lost placeholder, gained real panels
Admin unlock	Google login plus hard-coded developer emails (old/src/App.tsx:499-500)	?admin=true password prompt, sessionStorage flags (cur/src/App.tsx:631-663)	Replaced
Global margins/settings	Does not exist	Saved to D1 via /api/settings/:key, mirrored to localStorage (cur/src/App.tsx:1568-1582, cur/worker/index.ts:372-396)	New
Product import/export	Does not exist	JSON import/export, admin only (cur/src/App.tsx:1538-1566, cur/src/App.tsx:1586-1590)	New
Internet data import/export	Does not exist	JSON bundle of internet master devices and products (cur/src/App.tsx:1591-1620	New

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3. Calculation engine diff

File sizes: old `src/utils/solarCalculator.ts` 421 lines; current 743 lines.

3.1 Public API

Function	Old signature	Current signature	Cha
`calculateUserNeeds`	`(devices: Device[]) => LoadAnalysis` (`old/src/utils/solarCalculator.`)	Same (`cur/src/utils/solarCalculator.`)	Non
`simulateHourlySoC`	`(hourlyLoad, actualDailyYield, usableBatteryWh, maxChargeW, ccType, location)` (`old/src/utils/solarCalculator.` 68)	Same (`cur/src/utils/solarCalculator.` 68)	Non
`getLoadSheddingAdvice`	`(devices, deficit)` (`old/src/utils/solarCalculator.`)	Same (`cur/src/utils/solarCalculator.`)	Non
`buildCombinations`	`(location, devices, hardware{inverters,panels,batt batteryPreference="any", tolerance=20)` (`old/src/utils/solarCalculator.` 190)	`(location, devices, hardware{+powerstations}, batteryPreference="any", tolerance=20, products=[], margins={...})` (`cur/src/utils/solarCalculator.` 317)	Seve para two new opti inpu

3.2 Input handling

Aspect	Old	Current
Array guards	None; malformed API payload throws	`safeDevices`, `safeHardware`, `safeProducts` normalise missing arrays (`cur/src/utils/solarCalculator.ts:318-325`)
Missing PSH key	`LOCATION_PSH[location]` can be `undefined` (`old/src/utils/solarCalculator.ts:193`)	Falls back to 2.6 (`cur/src/utils/solarCalculator.ts:329`)

3.3 Search space

Loop	Old	Current
Inverter × panel × battery	Present (`old/src/utils/solarCalculator.ts:198-415`)	Present, same nested structure (`cur/src/utils/solarCalculator.ts:527-730`)
Pre-configured kits	None	Iterates `products` where `type === "combination"`, skips non-solar kits, checks inverter surge, battery and panel fit, applies product margin (`cur/src/utils/solarCalculator.ts:338-`

		438)
Powerstation	None	Iterates `hardware.powerstations`, checks capacity, output, PV input, applies powerstation margin and a flat panel price of 95,000 (`cur/src/utils/solarCalculator.ts:442-521`, `cur/src/utils/solarCalculator.ts:498-500`)

3.4 Status logic

Aspect	Old	Current
Deficit used for percentage	`sim.finalDeficitWh / total_daily_wh` (`old/src/utils/solarCalculator.ts:368-369`)	`getEffectiveDeficitWh = max(totalDaily - dailyYield, simDeficit)` (`cur/src/utils/solarCalculator.ts:300-307`, used at `cur/src/utils/solarCalculator.ts:680-682`)
Optimal	`sim.passed` (`old/src/utils/solarCalculator.ts:371-374`)	`sim.passed && effectiveDeficitWh === 0` (`cur/src/utils/solarCalculator.ts:690-693`)
Conditional	`deficitPercentage <= tolerance` (`old/src/utils/solarCalculator.ts:375-380`)	Same threshold on effective deficit (`cur/src/utils/solarCalculator.ts:694-698`)
High Risk	Kept and pushed with a warning (`old/src/utils/solarCalculator.ts:381-386`, `old/src/utils/solarCalculator.ts:393-411`)	Computed, logged, then discarded (`cur/src/utils/solarCalculator.ts:699-709`)
Status helper	None	`getStatusFromSimulation` exists but is never called (`cur/src/utils/solarCalculator.ts:208-219`)

The High Risk drop is the largest behavioural change in the engine. Old results included a warned, priced High Risk option; current results simply omit it. This interacts badly with the UI that still filters and sorts High Risk (`cur/src/App.tsx:2679`, `cur/src/App.tsx:2725`).

3.5 Battery compatibility

Aspect	Old	Current
Voltage check	Battery voltage must divide system DC voltage exactly, so 12.8 V lithium fails on a 12 V system (`old/src/utils/solarCalculator.ts:237-244`)	`getNominalBatteryVoltage` maps 12.8 V lithium to 12 V and `isBatteryVoltageCompatible` allows a 20 % deviation plus a modulo system match (`cur/src/utils/solarCalculator.ts:184-201`, used at `cur/src/utils/solarCalculator.ts:563-572`)
Effect	12.8 V lithium often rejected on 12 V systems	12.8 V lithium accepted on 12 V systems

3.6 Pricing

Aspect	Old	Current
Inverter	`inv.price * numUnits` (`old/src/utils/solarCalculator.ts:390`)	`applyMargin(inv.price * numUnits, margins.inverter)` (`cur/src/utils/solarCalculator.ts:712`)
Panel	`panel.price * totalPanels` (`old/src/utils/solarCalculator.ts:389`)	`applyMargin(panel.price * totalPanels, margins.panel)` (`cur/src/utils/solarCalculator.ts:711`)
Battery	`bat.price * totalBatteries` (`old/src/utils/solarCalculator.ts:283`)	`applyMargin(bat.price * totalBatteries, margins.battery)` (`cur/src/utils/solarCalculator.ts:612`)
Powerstation	N/A	`applyMargin(ps.price, margins.powerstation)` plus `applyMargin(panelEquivalentCount * 95000, margins.panel)` (`cur/src/utils/solarCalculator.ts:498-500`)
Kits	N/A	Product price and component prices each get `margins.product` (`cur/src/utils/solarCalculator.ts:409-417`)

The 95,000 hard-coded panel price at `cur/src/utils/solarCalculator.ts:500` is a magic number with no constant or comment.

3.7 Constants

Item	Old	Current
Inverters	3 (`old/src/constants.ts:22-62`)	3, unchanged models (`cur/src/constants.ts:24-64`)
Panels	2 (`old/src/constants.ts:64-82`)	10 (`cur/src/constants.ts:66-78`)
Batteries	3 lead/lithium (`old/src/constants.ts:83-114`)	5 lithium at 12.8 V (`cur/src/constants.ts:79-86`)
Powerstations	None	5 (`cur/src/constants.ts:87-145`)
Location/irradiance/surge tables	Present (`old/src/constants.ts:3-21`)	Identical (`cur/src/constants.ts:3-23`)

3.8 Log format

Both engines return `allLogs: string[][]` (`old/src/utils/solarCalculator.ts:190`, `cur/src/utils/solarCalculator.ts:317`). Current adds kit and powerstation trace rows, margin traces, and the effective-deficit explanation. The UI log viewer and TXT export consume the same shape, so no migration is needed there.

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4. Data, persistence, and API diff

4.1 Backend shape

Aspect	Old	Current
Runtime	Express 4 on port 3000 (`old/server.ts:17-30`, `old/server.ts:384-390`)	Cloudflare Worker with static assets (`cur/worker/index.ts:426-453`)
Database	Supabase if `SUPABASE_URL` + service key, else SQLite `solar_sizer.db` via better-sqlite3 (`old/server.ts:17-72`)	D1 binding `DB`, database `solar-sizer-sdk2` (`cur/wrangler.jsonc:12-19`)
Schema	`users`, `profiles`, `results`, `hardware` (`old/supabase.sql`)	`devices_master`, `hardware`, `products`, `settings` (`cur/migrations/0001_init.sql:1-39`, mirrored at `cur/worker/index.ts:98-134`)
Seeding	None; tables assumed populated by users	Auto-seeds on first boot: 25 devices, 23 products, 38 hardware rows (`cur/worker/index.ts:141-193`, `cur/worker/seed-data.ts:1-311`)
Health check	`/healthz` (`old/server.ts:79-84`)	`/healthz` before boot (`cur/worker/index.ts:430-432`), `run_worker_first` includes it (`cur/wrangler.jsonc:10`)
SPA fallback	Express static + `*` route (`old/server.ts:386-390`)	`not_found_handling: single-page-application` (`cur/wrangler.jsonc:9`)

4.2 Auth and session

Endpoint / behaviour	Old	Current
`GET /api/auth/user`	Returns session user (`old/server.ts:106-108`)	Always `{ user: null }` (`cur/worker/index.ts:238-241`)
`POST /api/auth/logout`	Clears session (`old/server.ts:110-113`)	Returns `{ success: true }` (`cur/worker/index.ts:243-245`)
`GET /api/auth/google/url`	Builds Google OAuth URL (`old/server.ts:116-139`)	Missing; 404
`GET /api/auth/google/callbac`	Exchanges code, upserts user, sets session (`old/server.ts:141-239`)	Missing; 404
Session storage	Signed cookie session (`old/server.ts:87-99`)	None
Auth UI	Google button, avatar, sign out (`old/src/components/Auth.tsx:19215`)	Renders `null` and calls `onUserChange(null)` (`cur/src/components/Auth.tsx:118`)

4.3 User data endpoints

Endpoint	Old	Current
`GET /api/user/data`	Returns the signed-in user's profiles, results, hardware (`old/server.ts:241-267`)	Returns empty arrays for everyone (`cur/worker/index.ts:247-249`)
`POST /api/user/profiles`	Upsert, auth required (`old/server.ts:269-298`)	Returns `{ success: true }`, discards body (`cur/worker/index.ts:251-256`)
`POST /api/user/results`	Upsert, auth required (`old/server.ts:300-325`)	Returns `{ success: true }`, discards body (`cur/worker/index.ts:251-256`)
	Upsert, auth required (	Returns `{ success: true }`, discards

`POST /api/user/hardware`	`upsert, addr required` (`old/server.ts:327-351`)	body (`cur/worker/index.ts:251-256`)
`DELETE /api/user/:type/:id`	Deletes from the user's table (`old/server.ts:353-376`)	Returns `{ success: true }` (`cur/worker/index.ts:258-260`)

The client no longer calls these endpoints. The stubs exist only to keep the old contract alive for future OAuth work, as the comment at `cur/worker/index.ts:238` states.

4.4 Catalog endpoints

Endpoint	Old	Current
`GET /api/devices`	Does not exist	Returns `devices_master`, tags parsed (`cur/worker/index.ts:262-265`, `cur/worker/index.ts:75-80`)
`POST /api/devices`	Does not exist	Admin upsert (`cur/worker/index.ts:267-287`)
`DELETE /api/devices/:id`	Does not exist	Admin delete (`cur/worker/index.ts:289-293`)
`GET /api/hardware`	Does not exist	Returns `user_id = 'system'` rows (`cur/worker/index.ts:295-298`)
`POST /api/hardware`	Does not exist	Admin upsert (`cur/worker/index.ts:300-320`)
`DELETE /api/hardware/:id`	Does not exist	Admin delete (`cur/worker/index.ts:322-326`)
`GET /api/products`	Does not exist	Products merged with tagged hardware as `standalone` products, optional `?tag=` (`cur/worker/index.ts:328-337`, adapter at `cur/worker/index.ts:210-220`)
`POST /api/products`	Does not exist	Admin upsert, enriches `combination_data` (`cur/worker/index.ts:339-364`, `cur/worker/enrich.ts:15-91`)
`DELETE /api/products/:id`	Does not exist	Admin delete (`cur/worker/index.ts:366-370`)
`GET/POST /api/settings/:key`	Does not exist	Public read, admin write (`cur/worker/index.ts:372-396`)
`POST /api/calculate`	Does not exist; calculation ran in the browser	Server-side calculation using the same TS engine and D1 products (`cur/worker/index.ts:398-421`)
`POST /api/checkout`	Does not exist	Called by the working-tree Buy button (`cur/src/App.tsx:2159`). Creates a Bachs checkout session (`cur/worker/index.ts:399-448`), returns `{ url }`, 503 until `BACHS_API_KEY` is set. Catalog ids are priced from D1; custom ids trust the client amount.

4.5 Client storage keys

Key	Old	Current
`ss_inverters`	Yes (`old/src/App.tsx:562`)	Yes (`cur/src/App.tsx:689`)
`ss_panels`	Yes (`old/src/App.tsx:572`)	Yes (`cur/src/App.tsx:699`)

ss_batteries	Yes (old/src/App.tsx:577)	Yes (cur/src/App.tsx:704)
ss_profiles	Yes (old/src/App.tsx:663)	Yes (cur/src/App.tsx:960)
ss_internal_logs	Yes (old/src/App.tsx:584)	Yes (cur/src/App.tsx:716, cur/src/App.tsx:944)
ss_has_generator	Yes (old/src/App.tsx:605)	Yes (cur/src/App.tsx:831-837)
ss_powerstations	No	Yes (cur/src/App.tsx:709, cur/src/App.tsx:959)
ss_results	No; results came from the server	Yes (cur/src/App.tsx:610, cur/src/App.tsx:1018)
ss_margins	No	Yes (cur/src/App.tsx:961)
ss_pdf_settings	No	Yes (cur/src/App.tsx:727-729, cur/src/App.tsx:1577)
ss_admin_unlocked	No	Yes, sessionStorage (cur/src/App.tsx:632, cur/src/App.tsx:651)
ss_admin_key	No	Yes, raw password in sessionStorage (cur/src/App.tsx:652)

4.6 Data ownership and merge behaviour

Old:

- Profiles and results hydrate from the server when signed in, else from localStorage for profiles and an empty results list (old/src/App.tsx:502-557).
- Hardware hydrates from localStorage and merges server hardware by id, preserving local-only entries (old/src/App.tsx:511-549).

Current:

- All user data is local (cur/src/App.tsx:610-612, cur/src/App.tsx:964-989).
- Global catalog hardware replaces local hardware wholesale whenever the global list is non-empty (cur/src/App.tsx:674-683), then the persistence effect writes the replacement back (cur/src/App.tsx:955-962).
- No merge by id, no conflict resolution, no user scoping.

This is the single largest data-integrity risk in the current tree.

---

5. Exports and imports diff

5.1 PDF exports

Export	Old	Current	Status
Single result PDF	Defined old/src/App.tsx:914, never called	Defined cur/src/App.tsx:1171, never called cur/src/App.tsx:1224-1289; adds Peak Load row, accepts savedDevices so a saved analysis prints its own device snapshot (cur/src/App.tsx:1225-1227), uses saved devices at call sites (	Dead in both
Results PDF	old/src/App.tsx:967-1029; surge, nighttime, daily, count, device list, systems table		Improved, one bad metric

		cur/src/App.tsx:3851, cur/src/App.tsx:3892)	
Profile PDF	Defined old/src/App.tsx:1031, never called; profiles tab printed inline (old/src/App.tsx:2404-2428)	Defined cur/src/App.tsx:1291, never called; no inline profile print	Lost
Hardware database PDF	old/src/App.tsx:1059-1094; three tables	cur/src/App.tsx:1319-1363; adds a powerstation table (cur/src/App.tsx:1351-1357)	Improved
Comparison PDF	Built inside ComparisonModal (old/src/App.tsx:212-247); always has a Generator column; correct initialCost	Hoisted to exportComparisonToPDF (cur/src/App.tsx:1101-1167); Generator column conditional but rows unconditional; initialCost = 0; two old rows (Daily Usage, Fuel Price) replaced by Reliability (cur/src/App.tsx:1135-1142)	Degraded
Quote PDF	Fixed title "SOLARSIZER PRO QUOTE", fixed QT- prefix, specs first, costs second, no Units column, fixed footer (old/src/App.tsx:1445-1524)	Editable title/prefix/footers (cur/src/App.tsx:2039-2040, cur/src/App.tsx:2060, cur/src/App.tsx:2146-2147), costs first with Units column (cur/src/App.tsx:2077-2093), specs second (cur/src/App.tsx:2103-2118), wiring guide third (cur/src/App.tsx:2128-2140), VAT footer unchanged (cur/src/App.tsx:2092)	Improved
Usage profile PDF	Does not exist	Generated on demand and automatically with every quote (cur/src/App.tsx:1922-2032, called at cur/src/App.tsx:2152); overnight ranges fixed in working tree (cur/src/App.tsx:1973)	New, forced side effect

5.2 JSON, TXT, Markdown

Export	Old	Current	Status
Results JSON	metadata{generated,region,batteryPreference , analysis, devices, systems (old/src/App.tsx:1096-1109)	Identical (cur/src/App.tsx:1365 1376)	Same, still no powerstation key (not applicable to old)
Hardware JSON	{ inverters, panels, batteries } (old/src/App.tsx:1111-1115)	{ inverters, panels, batteries } (cur/src/App.tsx:1380 1384)	Same code, now incomplete
Hardware Markdow	Three sections (old/src/App.tsx:1304-1341)	Three sections (cur/src/App.tsx:1726 1761), no powerstation section	Same code, now incomplete
Profiles	old/src/App.tsx:1224-1228	cur/src/App.tsx:1646	Same

JSON	old/src/App.tsx:1224-1226	1650	Same
Full logs TXT	old/src/App.tsx:1250-1270	cur/src/App.tsx:1672-1692	Same
Single log TXT	old/src/App.tsx:1272-1301	cur/src/App.tsx:1694-1723	Same
Products JSON	Does not exist	cur/src/App.tsx:1586-1590	New
Internet bundle JSON	Does not exist	{ masterDevices, products } filtered by internet tag (cur/src/App.tsx:1591-1596)	New
Usage profile data	Does not exist	Not exported as JSON, only PDF	Gap

5.3 Imports

Import	Old	Current	Status
Hardware database	Accepts inverters, panels, batteries, sanitises numbers and ids (old/src/App.tsx:1117-1222)	Same plus powerstations (cur/src/App.tsx:1386-1506, powerstation branch at cur/src/App.tsx:1472-1483)	Improved
Profiles	Merge by id/name (old/src/App.tsx:1229-1248)	Same (cur/src/App.tsx:1651-1670)	Same
Master devices	Does not exist	Admin JSON import, upserts through SDK (cur/src/App.tsx:1508-1536)	New
Products	Does not exist	Admin JSON import, upserts through SDK (cur/src/App.tsx:1538-1566)	New
Internet bundle	Does not exist	Imports master devices then products (cur/src/App.tsx:1599-1644)	New

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6. UI and UX diff

6.1 Global shell

Element	Old	Current
Header	Logo, nav, Save Profile, auth widget (old/src/App.tsx:1551-1615)	Logo, nav, Save Profile, no auth widget (cur/src/App.tsx:2208-2278)
Nav visibility	Profiles/results/database hidden until sign-in; logs hidden until developer email (old/src/App.tsx:1568-1597)	Everything visible except logs, which need admin unlock (cur/src/App.tsx:2220-2264)
Compact mode	None	Header and footer suppressed with ?compact=true (cur/src/App.tsx:2209, cur/src/App.tsx:617-627)
Footer	None in old (old/src/App.tsx ends after	Marketing footer, link columns, newsletter input

	main content)	(`cur/src/App.tsx:4053-4100`)
Theme	Stone/emerald, same palette	Same palette, light UI made the main interface at HEAD `6ccd8fe`

6.2 Calculator screen

Element	Old	Current
Empty state	No dedicated empty state; results column blank	Empty state panel with `min-h-[420px]` so it stays at the top (`cur/src/App.tsx:2554`; HEAD `863310d`; working tree retains it)
Device card	Name, watts, qty, category, ranges, edit/delete (`old/src/App.tsx:1618-1916`)	Same plus master-device chips (`cur/src/App.tsx:2280-2590`)
Range editor	Start/end selects, add/remove (`old/src/App.tsx:1395-1412`)	Same (`cur/src/App.tsx:1817-1834`)
Results header	Count, comparison bar (`old/src/App.tsx:1916-1940`)	Count, All/Kits/Powerstations filter, pagination (`cur/src/App.tsx:2603-2630`)
System card	Status chip, price, specs, save/compare/quote buttons (`old/src/App.tsx:1940-2148`)	Same plus High Risk dead branches and "Included" price fallbacks (`cur/src/App.tsx:2700-2910`)
Detail modal	Three-column specs, conditional bridge button (`old/src/App.tsx:2892-2960`)	Redesigned banner, three cards, Buy button, bridge for Conditional only (working tree `cur/src/App.tsx:4101-4340`)
Comparison modal	Side-by-side metrics, generator settings, PDF (`old/src/App.tsx:171-453`)	Same, PDF delegated to App (`cur/src/App.tsx:313-565`)

6.3 Products and Internet tabs

Element	Old	Current
Product card	None	Image/icon, name, description, price, tags, specs, Buy (working tree), admin edit/delete (`cur/src/App.tsx:2990-3070`)
Tag filters	None	Flagship, Solar, Internet, Powerstation, All (`cur/src/App.tsx:2950-2990`)
Admin toolbar	None	Import products, export products, add product (`cur/src/App.tsx:2918-2938`)
Internet page	None	Hero copy, import/export, master device list, product list (`cur/src/App.tsx:3074-3190`)
Empty states	None	Present for products and internet lists

6.4 Profiles, results, database, logs

Element	Old	Current
Profiles	Search, load, PDF, delete (`old/src/App.tsx:2365-2498`)	Search, load, delete, JSON export/import (`cur/src/App.tsx:3612-3736`)
Results	Server list, view, PDF, delete (`old/src/App.tsx:2498-2831`)	Local list, view, PDF, delete, comparison view (`cur/src/App.tsx:3736-4053`)
Database	Three sections, add/edit/duplicate/delete, import/export (`old/src/App.tsx:2148-2281`)	Four sections, plus master devices and margins/PDF panels behind admin (`cur/src/App.tsx:3190-3515`)
Logs	Developer only, expandable traces (`old/src/App.tsx:2281-2365`)	Admin only, duplicate suppression, clear button, per-log TXT (`cur/src/App.tsx:3515-3612`)

6.5 Accessibility and mobile

Both trees use the same patterns: icon buttons with `title` attributes but no `aria-label` in many places, modals without focus traps, and no mobile navigation replacement for `hidden md:flex`. Current does not regress these, but it also does not fix them. New controls (Buy, admin imports) continue the icon-button pattern without accessible names (`cur/src/App.tsx:4311-4318`).

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7. Architecture and build diff

Aspect	Old	Current
Frontend	Vite 6 + React 19 + Tailwind 4 + Motion + jsPDF (`old/package.json:17-33`)	Same plus no Gemini/Supabase (`cur/package.json:19-30`)
Backend	Express + cookie-session + better-sqlite3 + Supabase (`old/package.json:18-29`)	Cloudflare Worker + D1 (`cur/wrangler.jsonc`, `cur/worker/index.ts`)
Dev command	`tsx server.ts` serves API and Vite middleware (`old/package.json:6`, `old/server.ts:384`)	`vite` only, `/api` proxied to `127.0.0.1:8787` (`cur/vite.config.ts:36-38`, `cur/package.json:6`)
Worker dev	N/A	`wrangler dev --local --port 8787` (`cur/package.json:7`, `cur/package.json:10-11`)
Build	`vite build` (`old/package.json:7`)	`vite build` plus manual vendor chunks for react, UI, PDF (`cur/vite.config.ts:8-28`)
Deploy	Not configured in repo	GitHub Actions to Cloudflare Workers, needs `CLOUDFLARE_API_TOKEN` (`cur/.github/workflows/deploy.yml:1-26`)
Database migrations	`supabase.sql` applied manually (`old/supabase.sql`)	`wrangler d1 migrations apply` with `migrations/0001_init.sql` (`cur/package.json:15`)
Seeding	None	`worker/seed-data.ts`, auto-applied on first request (`cur/worker/index.ts:141-193`)
Tests	None	`worker/enrich.test.ts` via `npm run test:worker` (`cur/package.json:14`)
Typing	`tsc --noEmit` (`old/package.json:8`)	Same (`cur/package.json:12`)
SDK	None	`src/sdk/index.ts` wraps axios calls for devices, products, hardware, settings, calculate (`cur/src/sdk/index.ts:1-88`)
CLI	None	`cli.ts` lists devices and runs a calculation against the local Worker (`cur/cli.ts:1-66`)
Docs	README only	README, `INTEGRATION_GUIDE.md`, `docs/agents/{domain,issue-tracker,triage-labels}.md`, `AGENTS.md`
Env	Gemini, App URL, session secret, Google OAuth, Supabase (`old/.env.example:1-21`)	`ADMIN_PASSWORD`, optional `APP_URL` (`cur/.env.example:1-7`)
Title	"My Google AI Studio App" (`old/index.html`)	"SolarSizer Pro — Intelligent Solar Sizing" (`cur/index.html`)

Two build-level consequences follow. First, `npm run dev` alone no longer serves the API, so a fresh clone

appears broken until the developer starts `dev:worker` as well. Second, the deployed Worker serves the SPA and the API from one origin, which removes CORS and cookie-domain problems but makes the Worker the single point of failure for both.

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8. Port recommendations

Ordered by value per unit of work. Each item names the file and the smallest change that closes the gap.

1. **Restore a real identity layer or remove the pretence.** Either port the Google OAuth flow and the per-user tables (`old/server.ts:101-376`, `old/supabase.sql`) to a Worker-compatible auth provider, or delete the no-op endpoints (`cur/worker/index.ts:238-260`) and the `Auth` stub (`cur/src/components/Auth.tsx`) and label the product as local-only. Keeping silent success responses is the worst option because callers believe data was saved.
2. **Fix the hardware merge before anything else ships.** Replace the wholesale assignment at `cur/src/App.tsx:680-683` with the old merge-by-id logic (`old/src/App.tsx:511-549`), and only fall back to global defaults when local storage is empty. Add a unit test for "local item survives a global refresh".
3. **Decide whether High Risk belongs in the product.** If yes, delete the `continue` at `cur/src/utils/solarCalculator.ts:706-709` and keep the existing UI branches. If no, delete the High Risk branches, filters, and sort keys (`cur/src/App.tsx:2679`, `cur/src/App.tsx:2725`, `cur/src/App.tsx:4148-4156`) so the UI does not advertise a status the engine cannot return. Do not leave both halves.
4. **Finish the checkout flow.** The working-tree Buy button posts product id, amount, and description (`cur/src/App.tsx:2155-2178`) and the Worker route exists (`cur/worker/index.ts:399-448`). It prices catalog ids from D1 but still trusts `amount` for custom systems and hardware-derived products. Re-price those server-side or create the session from a stored quote. Document `BACHS_API_KEY` in `.env.example`, read `?payment=success|cancelled` in the UI instead of dropping it, and hide the button while the key is unset (today it opens an alert).
5. **Repair the comparison PDF.** Restore `initialCost = hasGenerator ? 0 : selectedGen.price` (`old/src/App.tsx:206`), build rows conditionally with the header, and re-add the Daily Usage and Fuel Price rows if they matter to customers. Add one assertion-style check that every row length equals `headers.length`.
6. **Add powerstations to every hardware export.** Include `powerstations` in `exportHardwareDatabaseJSON` (`cur/src/App.tsx:1380-1384`) and add section 4 to the Markdown export (`cur/src/App.tsx:1726-1761`). The importer already accepts them, so this closes a round-trip data loss.
7. **Replace the single admin password with a scoped token or per-user role.** At minimum, add a Worker rate limit on `/api/admin/verify` (`cur/worker/index.ts:226-236`), stop storing the raw password in `sessionStorage` (`cur/src/App.tsx:652`), and validate catalog payloads server-side, including price and ids.
8. **Make the usage profile opt-in and verify its maths.** The quote currently downloads two files with no warning (`cur/src/App.tsx:2152`). Add a checkbox beside "Generate Quote". Keep the working-tree overnight fix (`cur/src/App.tsx:1973`) and add a test for a 20:00-02:00 range.

9. **Remove dead exports and helpers.** Delete `exportSingleResultToPDF` and `exportProfileToPDF` or wire them to visible buttons; delete `getStatusFromSimulation` (`cur/src/utils/solarCalculator.ts:208`) or use it in the main loop; move `buildProductDetails` and `buildMasterDeviceDetails` into their proper scope (`cur/src/App.tsx:221-300`).
10. **Make local development one command.** Add a `dev:all` script that runs Vite and Wrangler together, or proxy through Wrangler only. Document the two-terminal requirement in the README, and add a `dev` guard that fails with a clear message when `127.0.0.1:8787` is not listening.
11. **Reconsider the "Peak Load" label.** Either compute a true instantaneous peak (which requires per-device surge overlap, already available in `hourly_surge`) or rename the row to "Highest Hourly Load" so the report does not overstate what it measures (`cur/src/App.tsx:1246-1250`).
12. **Close the exports gap for kits and powerstations.** Results JSON already carries the systems, but hardware JSON/Markdown omit powerstations. Consider adding a `products` and `powerstations` block to the results export so a customer quote bundle can be reconstructed offline.

Appendix A: Endpoint inventory

Old (`old/server.ts`):

Method	Path	Auth	Line
GET	`/healthz`	No	79
GET	`/api/auth/user`	No	106
POST	`/api/auth/logout`	No	110
GET	`/api/auth/google/url`	No	116
GET	`/api/auth/google/callback`	No	141
GET	`/api/user/data`	Yes	241
POST	`/api/user/profiles`	Yes	269
POST	`/api/user/results`	Yes	300
POST	`/api/user/hardware`	Yes	327
DELETE	`/api/user/:type/:id`	Yes	353

Current (`cur/worker/index.ts`):

Method	Path	Auth	Line
GET	`/healthz`	No	430
POST	`/api/admin/verify`	Password body	226
GET	`/api/auth/user`	No, stub	239
POST	`/api/auth/logout`	No, stub	243
GET	`/api/user/data`	No, stub returns empty	247
POST	`/api/user/{profiles,results,hardware}`	No, stub discards	251
DELETE	`/api/user/*`	No, stub	258
GET	`/api/devices`	No	262
POST	`/api/devices`	Admin header	267
DELETE	`/api/devices/:id`	Admin header	289

GET	`/api/hardware`	No	295
POST	`/api/hardware`	Admin header	300
DELETE	`/api/hardware/:id`	Admin header	322
GET	`/api/products`	No	328
POST	`/api/products`	Admin header	339
DELETE	`/api/products/:id`	Admin header	366
GET/POST	`/api/settings/:key`	Read public, write admin	372
POST	`/api/calculate`	No	398

Appendix B: File inventory deltas

Added in current: `worker/` (index, enrich, enrich.test, seed-data), `migrations/0001_init.sql`, `wrangler.jsonc`, `worker-configuration.d.ts`, `src/sdk/index.ts`, `src/components/InteractiveBridge.tsx` (same as old), `cli.ts`, `INTEGRATION_GUIDE.md`, `.github/workflows/deploy.yml`, `docs/agents/`, `AGENTS.md`.

Removed from current: `server.ts`, `supabase.sql`, `.env.example` OAuth/Supabase/Gemini entries, `@google/genai`, `@supabase/supabase-js`, `better-sqlite3`, `cookie-session`, `express`, `dotenv`, `tsx`.

Modified in current: `src/App.tsx`, `src/utils/solarCalculator.ts`, `src/types.ts`, `src/constants.ts`, `src/components/Auth.tsx`, `vite.config.ts`, `index.html`, `package.json`, `metadata.json`.

Unchanged in current: `src/components/ErrorBoundary.tsx` (62 lines, byte-identical to old), `src/components/InteractiveBridge.tsx` (148 lines, byte-identical apart from line endings), `src/main.tsx`.